

Name

$$\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} - 2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\mathcal{W} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\begin{aligned} 0u_1 + 1u_2 &= 0 \\ 0u_1 + 0u_2 &= 0 \end{aligned}$$

$$\vec{u} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

■ For the ODE $y' = Ay$ with A below. Write the gen sol and the important part as $t \rightarrow \infty$.

In[]:= MatrixForm[Eigensystem[$\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$]]

Out[]:= MatrixForm=

$$\begin{pmatrix} 2 & 2 \\ (1, 0) & (0, 0) \end{pmatrix}$$

$$y_1 = e^{2t} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$y_2 = te^{2t} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + e^{2t} \vec{u}$$

$$y(t) = c_1 y_1(t) + c_2 y_2(t)$$

$$\approx c_2 y_2$$

1. For the ODE $y' = Ay$ with A below. Write the gen sol.

$$y(t) =$$

In[]:= MatrixForm[Eigensystem[$\begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}$]]

Out[]:= MatrixForm=

$$\begin{pmatrix} 1 & 1 \\ (1, 1) & (0, 0) \end{pmatrix}$$

■ For the ODE $y' = A y$ with A below. Write the gen sol and the important part as $t \rightarrow \infty$.

In[] := MatrixForm[Eigensystem[$\begin{pmatrix} 0 & -5 & -2 \\ 1 & 0 & 1 \\ -2 & 4 & 0 \end{pmatrix}$]]

Out[] := MatrixForm[

$\begin{pmatrix} 2 & -1 & -1 \\ (-1, 0, 1) & (-1, -1, 2) & (0, 0, 0) \end{pmatrix}$

$$y_1 = e^{2t} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$y_2 = e^{-t} \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} \quad y_3 = t y_2 + e^{-t} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

$$y(t) = c_1 y_1 + c_2 y_2 + c_3 y_3 \approx c_1 y_1$$

$$\left[A = (-1) \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right] \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} \quad \begin{bmatrix} 1 & -5 & -2 \\ 1 & 1 & 1 \\ -2 & 4 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$$

$\begin{matrix} u_1 = -1 \\ u_2 = 0 \\ u_3 = 0 \end{matrix}$
 $\sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

2. For the ODE $y' = A y$ with A below. Write the gen sol and the important part as $t \rightarrow \infty$.

$$y(t) = \quad \approx \quad$$

In[] := MatrixForm[Eigensystem[$\begin{pmatrix} 2 & 6 & -3 \\ -2 & -5 & 2 \\ -1 & -2 & -4 \end{pmatrix}$]]

Out[] := MatrixForm[

$\begin{pmatrix} -3 & -3 & -1 \\ (3, -2, 1) & (0, 0, 0) & (-2, 1, 0) \end{pmatrix}$